HINTS & SOLUTION

Q.1 (C)
mvr = nh/2p

$$v = \frac{2h}{2\pi m 4a} \qquad r \propto \frac{n^2}{z}$$

$$\frac{a}{x} = \frac{1}{4}$$

$$\frac{1}{2}mv^2 \Rightarrow \frac{1}{2}\frac{m \times h^2}{\pi^2 m^2 \times 16a_0^2} \qquad x = 4a$$

$$\text{K.C.} = \frac{h^2}{32m\pi^2 a_0^2} \qquad \text{Ans. (C)}$$

Q.2 (B)
$2\pi r_n = n\lambda$
$4 \longrightarrow 2$

$$\lambda_4 = \frac{2\pi r_4}{4} \qquad \lambda_2 = \frac{2\pi r_2}{2}$$

$$\lambda_4 - \lambda_2 = 2\pi\left[\frac{0.529(4)^2}{1\times 4} - \frac{0.529(2)^2}{2}\right]\text{Å}$$

$= 2\pi\,(0.529)\,(2)$
$= 4\pi r_1 = 6.64$ Å

Q.3 (C)
For distance of closest approach

$$\frac{1}{2}mV^2 = \frac{Kq_1q_2}{r_{min}}$$

for Zn nucleus $q_1 = 30\,e = 30 \times 1.6 \times 10^{-19}$ C

for charged particle $\left(\frac{q_2}{m}\right) = 4 \times 10^{10}$ col/kg

$$\frac{1}{2}V^2 = \frac{Kq_1\left(\frac{q_2}{m}\right)}{r_{min}}$$

$$r_{min} = \frac{2\times(30\times1.6\times10^{-19})\times(4\times10^{10})\times9\times10^9}{(3\times10^6)^2} = 3.84\text{ Å}$$

Q.4 (A)
$n_2 = 11$ higher state
$n_1 = 1$ lowest state
U.V. $= 11 - 1 = 10$
visible $= 11 - 2 = 9$

$$\text{remaining} = \frac{11\times10}{2} - 19 = 55 - 19$$

(z) I.R. = 36

Hence $z - (x + y) = 36 - (9 + 10) = 17$

Q.5 (D)

$4s^4 \qquad 3d^{20}\ 4p^{12}$

$4 + 12 + 20 = 36$

Q.6 (C)

$$T \propto \frac{n^3}{Z^2}$$

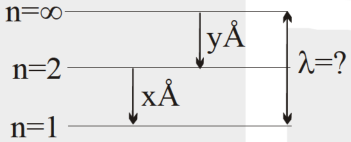
$$\frac{x}{y} = \frac{9}{3} \times \frac{2^2}{2^3}$$

$$y = \frac{2}{3}x \ \text{ sec}$$

Q.7 (C)

$$\frac{1}{\lambda} = \frac{1}{x\text{Å}} + \frac{1}{y\text{Å}}$$

$$\therefore \lambda = \frac{xy}{x+y}\text{Å}$$



Q.8 (D)

$$\sqrt{l(l+1)}\frac{h}{2\pi} = \sqrt{3}\frac{h}{\pi}$$

$\therefore l = 3$

So value of $n > 3$

$$\therefore \text{Angular momentum} = \frac{nh}{2\pi}(n\varepsilon I)$$

$$\text{So angular momentum} > \frac{3h}{2\pi}$$

$$\text{So } 2.\frac{h}{\pi}$$

Q.9 (C)

$$E = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34+8+9} \times 3}{12.4} = 1.6 \times 10^{-17}\ \text{J} \qquad 1\ \text{eV} = 1.6 \times 10^{-19}\ \text{J}$$

$$= 100\ \text{eV}$$

$$\lambda = \frac{h}{\sqrt{2mKE}} = \frac{12.25}{\sqrt{45.6}} = 1.8\ \text{Å}$$

Q.10 (A)

4 to 2 transition in He^+ ion correspond to 2 to 1 in H-atom i.e. lies in ultraviolet region

Q.11 (B)

$S_1 \rightarrow 3s$

$S_2 \rightarrow 3d$ for unielectronic species energy depends only on 'n'

$S_3 \rightarrow 4s$ $\therefore S_1 = S_2 = S_4 < S_3$

$S_4 \rightarrow 3p$

Q.12 (C)

Q.13 (A)

Q.14 (C)

$E = IE_1 + IE_2$

$= 24.6 + 13.6 \times (2)^2$

$= 79$ eV

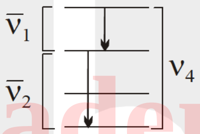
Q.15 (A) (B) (C)

Q.16 (C)

Q.17 (A) (B) (D)

$$\frac{n(n-1)}{2} = 6 \Rightarrow n = 4$$

(A) $E_4 = E_1 + E_2$



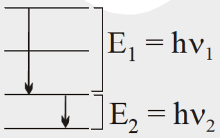
$$\frac{hc}{\lambda_4} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$\nu_4 = \nu_1 + \nu_2$

(B) $E = E_1 + E_2$

$h\nu = h\nu_1 + h\nu_2$

$\nu = \nu_1 + \nu_2$



(D) Longest wavelength $= \dfrac{1}{\lambda_{long}} = R_H\left[\dfrac{1}{3^2} - \dfrac{1}{4^2}\right]$ .... (i)

Shortest wavelength $= \dfrac{1}{\lambda_{short}} = R_H\left[\dfrac{1}{1^2} - \dfrac{1}{4^2}\right]$ ... (ii)

$$\frac{\lambda_{long}}{\lambda_{short}} = \frac{\left[1 - \frac{1}{16}\right]}{\left[\frac{1}{9} - \frac{1}{16}\right]} = \frac{\frac{15}{6}}{\left(\frac{7}{9\times 16}\right)}$$

$$= \frac{15}{16} \times \frac{9\times 16}{7} = \frac{135}{7}$$

Q.18 (C) (D)

Q.19 (B) (C) (D)

Q.20 (A)

Q.21 (B)

Q.22 (C)

Q.23 (B)

(i) $$\psi = \frac{2}{3}\left(\frac{Z}{3a_0}\right)^{3/2}(\sigma-1)(\sigma^2-8\sigma+12)\sigma e^{-\sigma/2}\cos\theta$$

$(\sigma-1)(\sigma^2 - 8\sigma + 12) = 0$

$\sigma = 1$ $\sigma = 2$ and $\sigma = 6$

Number of radial nodes = 3

$\psi(r) = Kr^l$ [Polynomial of degree $n - l - 1$] . $e^{-K/r}$

$$= K.\left(\frac{a_0\sigma}{2Z}\right)^l \text{[Polynomial] } e^{-K/r}$$

On comparing with the given equation

$l = 1$

Number of Angular node = 1

Total node = 3 + 1 = 4

(ii) $l = 1 \Rightarrow p_z$ orbital

(iii) For maximum distance

$\sigma = 6$

$$\frac{2Zr}{a_0} = 6$$

$$r = \frac{6a_0}{2Z} \Rightarrow r = \frac{3a_0}{Z}$$

Q.24 (A) Q,R (B) P, Q, R,S (C) P, Q, R (D) P, Q

Q.25 (A) P, (B) P,Q,S, (C) P, R (D) Q, S

Q.26 80

$$1s^2\ 2s^2\ \underset{\substack{\downarrow\\ \frac{2+1=3}{6e^-}}}{2p^6}\ \underset{\substack{\downarrow\\ \frac{3+0}{2e^-}}}{3s^2}\ 3p^6\ \underset{\substack{\downarrow\\ \frac{l=2}{10e^-}}}{3d^{10}}\ 4s^1$$

8 electron

$n + l = 3 = 8$ electron

$l = 2 = 10$ electron

$8 \times 10 = 80$

Q.27 8

We have $\rightarrow \Delta E = \frac{3}{4} \times 0.85$ eV

as energy = 0.6375 eV the photon will belong to brackett series
(as for brackett series $0.306 \le E \le 0.85$)

So $\frac{3}{4} \times 0.85 = 13.6 \left[\frac{1}{4^2} - \frac{1}{n^2}\right]$

$\Rightarrow n = 8$; Hence $x = 8$

Q.28 0100

Electron transit from higher level n to lower level 2.

$$6 = \frac{(n-2)(n-2+1)}{2}$$

$$n = 5$$

$\frac{1}{\lambda} = R_H \cdot z^2 \left(\frac{1}{{n_1}^2} - \frac{1}{{n_2}^2}\right)$ $\qquad n_1 = 1$ and $n_2 = 5$

$\therefore \frac{1}{\lambda} = \left(\frac{1}{96} \times 10^9\right) \times 1^2 x \left(\frac{1}{1^2} - \frac{1}{5^2}\right)$ $\qquad \therefore \lambda = 10^{-7}$ m = 100 nm

Q.29 0003

$A \longrightarrow 30$ W $\qquad E_A$ (photon) = 13.6 eV

$B \longrightarrow 40$ W $\qquad E_B$ (photon) = $13.6 \times 2^2 = 4 \times 13.6$ eV

For A $(13.6 \times 1.6 \times 10^{-19}) \times n_A = 30$

For B $(4 \times 13.6 \times 1.6 \times 10^{-19}) \times n_B = 40$

$\frac{n_A}{4n_B} = \frac{3}{4}$

$\Rightarrow \frac{n_A}{n_B} = \frac{3}{1}$

Q.30 300303